PROBLEM CL11

VTOL Propulsion Device in Hover: Nozzle Angle and Flow State

Original statement · CL11.pdf

Thrust vectoring | Bernoulli | Linear momentum | Energy

OFFICIAL PROBLEM STATEMENT

The propulsion device of a VTOL (vertical take-off and landing) aircraft includes a horizontal intake of cross section A_i and a thrust vectoring nozzle of variable angle α and exit cross section A_e. When the aircraft is hovering (suspended at a fixed point), the engine must provide a force $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$, such that the lift F_y equals the weight of the aircraft (Mg), while the thrust F_x is identically zero. The work per unit time on the air stream through the mobile blades is W and the amount of heat added per unit time is Q (including heat losses through the walls). For the analysis of the problem, one may introduce the following assumptions regarding the air motion:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform.
- The air around the engine casing is at rest with uniform pressure p_a.
- Upstream from the inlet the flow is inviscid and with constant density, such that $\rho_i = \rho_a$ and one may use Bernoulli's equation along any streamline to relate the ambient pressure p_a with that found at the entrance p_i.

With the above simplifications, write the equations that determine the values of U_i, U_e, p_i, ρ_e and α in terms of the known quantities ($p_a, Mg, \rho_a, A_i, A_e, \gamma, W$, and Q).

Source: CL11.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average); neglect $\bar{g}$; uniform properties at i and e; perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; ambient air at rest at p_a around the casing.

- Inlet: $\rho_i = \rho_a$ (constant density upstream) and Bernoulli from the still air to the inlet. Exit: free jet at p_a, density ρ_e.
- Geometry (figure): air enters along $-\bar{e}_x$ through the horizontal intake; the jet leaves downwards and backwards, $\bar{v}_e = -U_e(\sin\alpha\,\bar{e}_x + \cos\alpha\,\bar{e}_y)$, with α measured from the vertical.
- Hover: $F_x = 0, F_y = Mg$. Data: $[p_a,\ \rho_a,\ Mg,\ A_i,\ A_e,\ \gamma,\ \dot{W},\ \dot{Q}]$. Unknowns: $[U_i,\ U_e,\ p_i,\ \rho_e,\ \alpha]$: five equations are needed.

Symbol	Meaning	SI unit
A_i, A_e	intake and exit areas	m^2
U_i, U_e	intake and exit velocities	m/s
p_i, ρ_e	intake pressure, exit density	Pa, kg/m^3
α	nozzle angle from the vertical	rad
$\dot{W}, \dot{Q}$	power and heat given to the air	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the hovering aircraft (inertial while hovering); $\bar{e}_x$ horizontal, $\bar{e}_y$ vertical upwards.
- Surfaces: Σ_i (intake, $\bar{n} = +\bar{e}_x$, $\bar{v} = -U_i\bar{e}_x$), Σ_e (exit, $\bar{n} = \bar{v}_e/U_e$), Σ_w (fixed walls) and Σ_m (mobile blades).

$$\boxed{\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y = \int_{\Sigma_w,\Sigma_m} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}'\cdot\bar{n}\right] d\sigma} \qquad \text{(force of the air on the engine)}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.10)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_i	$\rho_a(-U_i)A_i$
Σ_e	$+\rho_e U_e A_e$
Σ_w, Σ_m	0

$$\boxed{\rho_a U_i A_i = \rho_e U_e A_e = \dot{m}} \qquad (1)$$

Entrance region (Notes Eq. 4.17, from the still ambient air to the inlet):

$$\boxed{p_a = p_i + \tfrac{1}{2}\rho_a U_i^2} \qquad (2)$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$\rho_a(-U_i\bar{e}_x)(-U_i)A_i = \rho_a U_i^2 A_i\,\bar{e}_x$	$-(p_i-p_a)A_i\,\bar{e}_x$	0
Σ_e	$-\rho_e U_e^2 A_e(\sin\alpha\,\bar{e}_x+\cos\alpha\,\bar{e}_y)$	0 (free jet)	0
$\Sigma_w,\ \Sigma_m$	0	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F} = -F_x\bar{e}_x - F_y\bar{e}_y$. x component with $F_x = 0$, (2) and (1): $\rho_a U_i^2 A_i - \rho_e U_e^2 A_e \sin\alpha = \frac{1}{2}\rho_a U_i^2 A_i$, i.e. $\frac{1}{2}\rho_a U_i^2 A_i = \rho_a U_i A_i\, U_e \sin\alpha$. y component with $F_y = Mg$: $-\rho_e U_e^2 A_e \cos\alpha = -Mg$.

$$\boxed{U_i = 2\,U_e\sin\alpha} \quad (3) \qquad \boxed{Mg = \rho_e U_e^2 A_e\cos\alpha = \rho_a U_i^2 A_i\,\frac{\cos\alpha}{2\sin\alpha}} \quad (4)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\tfrac{v^2}{2}\right)dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}\left(e_i+\frac{U_i^2}{2}\right)$	$+p_i U_i A_i$	0	0
Σ_e	$+\dot{m}\left(e_e+\frac{U_e^2}{2}\right)$	$-p_a U_e A_e$	0	0
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int_{\Sigma_m}\bar{q}\cdot\bar{n}\,d\sigma$
Σ_w	0	0	0	$-\int_{\Sigma_w}\bar{q}\cdot\bar{n}\,d\sigma$

Boxed terms: the work on Σ_m is $\dot{W}$; the heat through the walls plus the volume term $\int_{V_c} Q_R\,dV$ is $\dot{Q}$ (net, including the losses). With $h = e + p/\rho = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\rho_a U_i A_i \left[\left(\frac{\gamma}{\gamma-1}\frac{p_a}{\rho_e}+\frac{U_e^2}{2}\right)-\left(\frac{\gamma}{\gamma-1}\frac{p_i}{\rho_a}+\frac{U_i^2}{2}\right)\right]=\dot{W}+\dot{Q}} \quad (5)$$

(1) to (5) are five equations for U_i, U_e, p_i, ρ_e and α, which answers the question.

PHASE 4

Interpretation, Limits and Dimensional Check

- (3) says that the backward component of the jet momentum must cancel the intake momentum plus the suction force at the intake: because the intake is at a lower pressure than ambient (2), only half of $\rho_a U_i^2 A_i$ has to be balanced.
- (4): the lift is the vertical momentum flux of the jet. For $\alpha \to 0$ (vertical jet), (3) would require $U_i \to 0$: the intake momentum could not be balanced, so a hovering engine with a horizontal intake needs $\alpha > 0$.
- Dimensions (SI): $[\rho_e U_e^2 A_e] = \mathrm{N} = [Mg]$; $[\dot{m}\,\gamma p/((\gamma-1)\rho)] = \mathrm{W}$.
- Numerical illustration (hypothetical data, solved by Newton's method on the two equations left after eliminating U_i, p_i, ρ_e): $\rho_a = 1.2$ kg/m^3, $p_a = 101325$ Pa, $A_i = 2.5$ m^2, $A_e = 1.2$ m^2, $Mg = 80$ kN, $\dot{W}+\dot{Q} = 6$ MW, $\gamma = 1.4$ give $U_e = 212.22$ m/s, $\alpha = 71.85°$, $U_i = 403.32$ m/s, $p_i = 3727.19$ Pa, $\rho_e = 4.7511$ kg/m^3 (residuals below 10^{-6}).
- Agreement with the official solution: CL11.pdf obtains (1) to (5).